Electrostatic potential and Capacitance

1. The work done in bringing a unit positive charge from infinite distance to a point at distance x from a positive charge Q is W. Then the potential at that point is

1) $\frac{WQ}{x}$
2) W
3) $\frac{W}{x}$
4) WQ

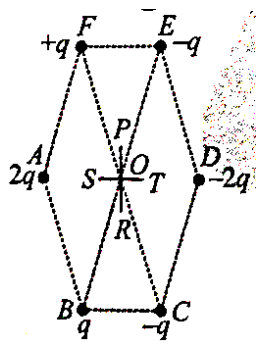
2. Electric charges of $+10\mu C, _5\mu C, -3\mu C$ and $+8\mu C$ are placed at the corners of a square of side $\sqrt{2}m$. The potential at the centre of the square is

1) 1.8 V
2) 1.8×10^{6} V
3) 1.8×10^{5} V
4) 1.8×10^{4} V

3. At a certain distance from a point charge, the field intensity is 500 V/m and the potential is - 3000 V. The distance of the charge and the magnitude of the charge respectively are

1) 6m and $6\mu C$
2) 4m and $2\mu C$
3) 6m and $4\mu C$
4) 6m and $2\mu C$

4. Six unit point charges are kept at the vertices of a regular hexagon of side L and centre O, as shown in the figure. Given that $K = \frac{q}{4\pi\varepsilon_0 L^2}$, which of the following statements is wrong?

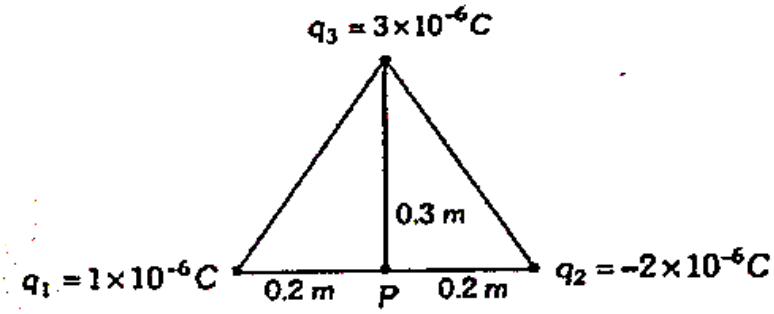


1) The electric field at O is 6 K along OD
2) The potential at O is zero
3) The potential at all points on the line PR is same
4) The potential at all points on the line ST is same

5. Two charges of 4 μC each are placed at the corners A and B of an equilateral triangle ABC of side length 0.2 m in air. The electric potential at C is $\left[\frac{1}{4\pi\varepsilon_0} = 9\times10^{9}\frac{N-m^2}{C^2}\right]$

1) $9\times10^{4}\,V$
2) $18\times10^{4}\,V$
3) $36\times10^{4}\,V$
4) $36\times10^{-4}\,V$

6. Figure shows a triangular array of three point charges. The electric potential V of these source charges at the midpoint P of the base of the triangle is

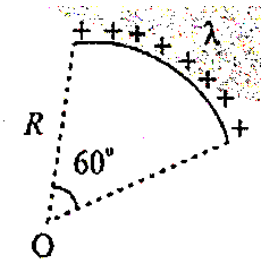


1) 55 kV
2) 45 kV
3) 63 kV
4) 49 kV

7. Electric field at a distance x from the origin is given as $E = \frac{100N-m^2/C}{x^2}$. The potential difference between the points situated at x = 10 m and x = 20 is:

1) 5 V
2) 10 V
3) 15 V
4) 4 V

8. Potential due to given charge A 29 F E +g P ST R D -29 distribution at centre (O) is



1) $\frac{\lambda}{6\varepsilon_0}$
2) $\frac{\lambda}{3\varepsilon_0 R}$
3) $\frac{\lambda}{12\varepsilon_0}$
4) $\frac{\lambda}{9\varepsilon_0}$

9. A non-conducting ring is of radius 0.5 m. 1.11×10^{-10} coulombs charge is non-uniformly distributed over the circumference of ring which produces electric field E around itself. If $\ell=0$ is the centre of the ring then the value of $\int_{\ell-\infty}^{\ell=0}-\vec{E}.d\ell$ is:

1) 2 V
2) – 2 V
3) – 1 V
4) zero

10. A rod of length L lies along the x -axis with its left end at the origin. It has charge Q uniformly distributed. Find the electric potential at point x = 2L

1) $\frac{KQ}{L}$
2) $\frac{KQ\log_e 2}{L}$
3) $\frac{2KQ}{L}$
4) $\frac{KQ}{L}\log 4$

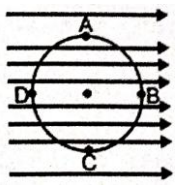
11. The electric potential in a certain region is expressed by $V=6x-8xy^2-8y+6yz-4z^2$ volts. The magnitude of the force acting on a charge of 2 C situated at the origin will be:

1) 2 N
2) 6 N
3) 8 N
4) 20 N

12. Electric potential at any point is $5x+3y+\sqrt{15}z,$ then the magnitude of electric field is

1) $3\sqrt{2}$
2) $4\sqrt{2}$
3) $5\sqrt{2}$
4) 7

13. A circle of radius R is drawn in a uniform electric field E as shown in the fig. V_A, V_B, V_C and V_D are respectively the potential of points A, B, C and D on the circle, then:

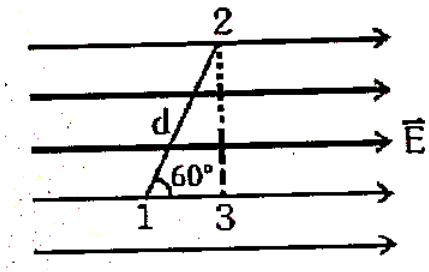


1) $V_A > V_C, V_B = V_D$
2) $V_A < V_C, V_B = V_D$
3) $V_A = V_C, V_B < V_D$
4) $V_A = V_C, V_B > V_D$

14. The electric potential V at any point O (x, y, z all in metres) in space is given by $V = 4x^2$ volt. The electric field at the point (1m, 0, 2m) in volt/metre is

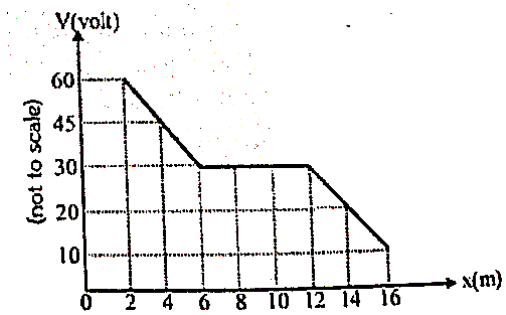
1) 8 along negative X-axis
2) 8 along positive X-axis
3) 16 along negative X-axis
4) 16 along positive Z-axis

15. The electric field $\vec{E}$ is constant in both magnitude and direction. Consider a path of length *d* at an angle $\theta = 60^0$ with respect to field lines as shown in figure. The potential difference between points 1 and 2 is



1) $\dfrac{E}{d\sin 60^0}$
2) $Ed\cos 60^0$
3) $\dfrac{Ed}{\cos 60^0}$
4) $\dfrac{E}{d}\sin 60^0$

16. The variation of potential with distance x from a fixed point is shown in figure. The electric field at x = 13 m is



1) 7.5 volt/meter
2) – 7.5 volt/meter
3) 5 volt/meter
4) – volt/meter

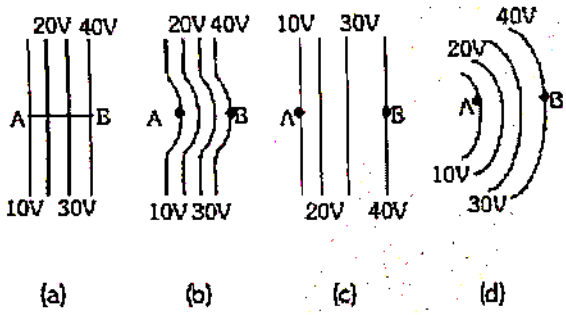
17. A uniform electric field having a magnitude E_0 and direction along the positive X-axis exists. If the potential V is zero at x = 0, then its value at X = + x will be.

1) $V(x) = +xE_0$
2) $V(x) = -xE_0$
3) $V(x) = +x^2E_0$
4) $V(x) = -x^2E_0$

18. A uniform electric field pointing in positive x - direction exists in a region. Let A be the origin, B be the point on the x - axis at x = +1 cm and C be the point on the y-axis at y = +1 cm. Then the potentials at the points A, B and C satisfy:

1) $V_A < V_B$
2) $V_A > V_B$
3) $V_A < V_C$
4) $V_A > V_C$

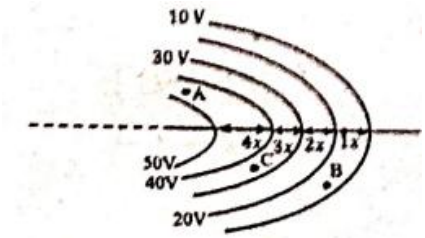
19. The diagrams below show regions of equipotential:



A positive charge is moved from A to B in each diagram.

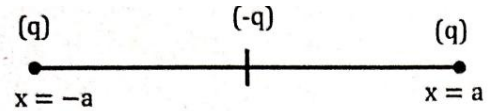
1) I all the four cases the work done is the same
2) Minimum work is required to move q in figure (a)
3) Maximum work is required to move q in figure (b)
4) Maximum work is required to move q in figure (c)

20. The figure shows lines of constant potential in a region in which electric field is present. The values of the potential are written in brackets, Of the points A, B and C, the magnitude of the electric field is greatest at the point:



1) A
2) B
3) C
4) unpredictable

21. Three charges are placed along x -axis at x = -a, x = 0 and x = a as shown in the figure. The potential energy of the system is



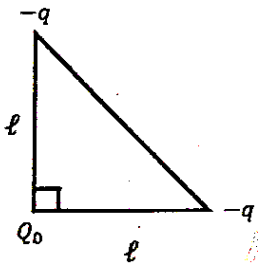
1) $-(\frac{1}{4\pi\varepsilon_0})\frac{q^2}{a}$

2) $-(\frac{1}{4\pi\varepsilon_0})\frac{3q^2}{2a}$

3) $+(\frac{1}{4\pi\varepsilon_0})\frac{q^2}{a}$

4) $+\left(\frac{1}{4\pi\varepsilon_0}\right)\frac{3q^2}{a}$

22. Three charges $Q_0 - q$ and $-q$ are placed at the vertices of an isosceles right angle triangle as in the figure. The net electrostatic potential energy is zero if Q_0 is equal to:



1) $\frac{q}{4}$

2) $\frac{2q}{\sqrt{32}}$

3) $\sqrt{2q}$

4) $\frac{q}{2\sqrt{2}}$

23. Charges 5μC and 10 μC are placed 1 m apart. Work done to bring these charges at a distance 0.5 m from each other is:

1) 9×10^4 J

2) 18×10^4 J

3) 45×10^{-2} J

4) 9×10^{-1} J

24. Four equal point charge Q each are placed in the xy-plane at (0, 2), (4, 2), (4, -2) and (0, -2). The work required to put a fifth charge Q at the origin of the coordinate system will be

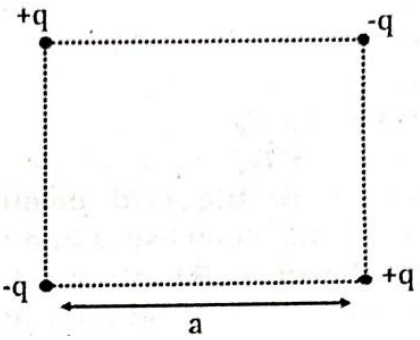
1) $\frac{Q^2}{4\pi\varepsilon_0}\left(1+\frac{1}{\sqrt{3}}\right)$

2) $\frac{Q^2}{4\pi\varepsilon_0}\left(1+\frac{1}{\sqrt{5}}\right)$

3) $\frac{Q^2}{4\pi\varepsilon_0}$

4) $\frac{Q^2}{2\sqrt{2}\pi\varepsilon_0}$

25. The work required to put the four charges together at the corners of a square of side 'a' as shown below is-



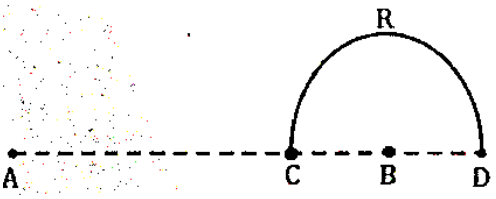
1) $\frac{Kq^2}{a}\left(4-\sqrt{2}\right)$

2) $\frac{Kq^2}{a}\left(\sqrt{2}-4\right)$

3) $\frac{Kq^2}{a}\frac{1}{(\sqrt{2}-4)}$

4) Zero

26. Charges +q and -q are placed at points A and B respectively which are a distance 2L apart, C is the midpoint between A and B. The work done in moving a charge +Q along the semicircle CRD is



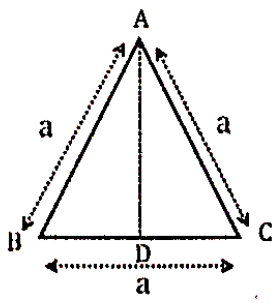
1) $\frac{qQ}{4\pi\varepsilon_0 L}$

2) $\frac{qQ}{2\pi\varepsilon_0 L}$

3) $\frac{qQ}{6\pi\varepsilon_0 L}$

4) $-\frac{qQ}{6\pi\varepsilon_0 L}$

27. At the corners of an equilateral triangle of side a (1 meter), three-point charges are placed (each of 0.1 C). If this system is supplied energy at the rate of 1 kW, then calculate the time required to move on charge at point A to midpoint of line joining BC.



1) 50 h

2) 60 h

3) 48 h
4) 54 h

28. Two particles each charged with a charge + q are clamped on the y axis at the points (0, a) and (0, -a). If a positively charged particle of charge q_o and mass m is projected towards the left along the x-axis from a point at a large distance on the right of the origin with a velocity $\sqrt{\frac{Kqq_0}{ma}}$, at what distance from origin will it come to rest

1) $\sqrt{15}a$
2) $4a$
3) $\sqrt{7}\,a$
4) $\sqrt{8}\,a$

29. If 'n' identical water drops assumed spherical each charged to a potential energy U coalesce to a single drop, the potential energy of the single drop is (Assume that drops are uniformly charged):

1) $n^{1/3}U$
2) $n^{2/3}U$
3) $n^{4/3}U$
4) $n^{5/3}U$

30. A solid sphere of radius R has total charge Q, uniformly distributed. Find the minimum work done to decrease the radius of the sphere to $\frac{R}{2}$.

1) $\frac{3}{5}\frac{KQ^2}{R}$
2) $\frac{6}{5}\frac{KQ^2}{R}$
3) $\frac{2}{5}\frac{KQ^2}{R}$
4) $\frac{1}{5}\frac{KQ^2}{R}$

Answer Key

(1)	2	**(16)**	3
(2)	3	**(17)**	2
(3)	4	**(18)**	2
(4)	4	**(19)**	1
(5)	3	**(20)**	2
(6)	2	**(21)**	2
(7)	1	**(22)**	4
(8)	3	**(23)**	3
(9)	1	**(24)**	2
(10)	2	**(25)**	2
(11)	4	**(26)**	4
(12)	4	**(27)**	1
(13)	3	**(28)**	1
(14)	1	**(29)**	4
(15)	2	**(30)**	1